A finite-speed return bridge: cuts and midpoint crossover

A velocity-resolved telegraph bridge gives one consistent random trajectory at every cut resolution. Its midpoint has a discrete probability mass, while the action error of sampled polygons vanishes at a controlled rate. The construction isolates velocity memory from the introduction of fresh noise at each cut.

1. Model and endpoint convention

Fix mass $m > 0$, speed $0 < u < c$, reversal rate $\lambda > 0$ and duration $T > 0$. In an inertial frame set $X_0 = 0$, $V_0 = u$ and

$$V_t = u(-1)^{N_t}, \qquad X_t = \int_0^t V_s\, ds,$$

where N is a homogeneous Poisson process of rate λ. Velocity is right-continuous. The rate is a supplied classical stochastic clock. This reduced model describes impulsive reversals; momentum receivers require the extra interaction bookkeeping of the physical-cut model (C030).

We construct the bridge to $(X_T, V_T) = (0, -u)$ by disintegrating switch-time densities at the interior position $X_T = 0$. The explicit simplex construction below fixes the version of this zero-probability conditioning. Before conditioning, the zero-switch event has probability $e^{-\lambda T}$ and lies at $(uT, +u)$; the chosen terminal velocity selects odd positive switch counts.

The free kinetic functional is $S[X] = (m/2)\int_0^T V_t^2 dt$, in action units. We compare it with the sampled polygon and with the zero-position chord. The chord matches position endpoints; its velocity endpoints differ from the bridge. All comparisons below name this positional endpoint convention.

2. Exact conditional path law

Write $N_T = 2K + 1$ and $z = \lambda T$. The bridge count law is

$$\boxed{\Pr(K = k \mid \text{bridge}) = w_k = \frac{(z/2)^{2k}}{(k!)^2 I_0(z)}, \qquad k = 0, 1, \ldots,}$$

where $I_0(z) = \sum_{k\geq 0}(z/2)^{2k}/(k!)^2$. Given $K = k$, independently choose uniform simplex vectors $(P_0, \ldots, P_k)$ and $(M_0, \ldots, M_k)$, each summing to $T/2$. Follow velocities $+u, -u, +u, -u, \ldots$ for durations $P_0, M_0, P_1, M_1, \ldots, P_k, M_k$. For $k = 0$ each simplex is a singleton. This specifies the entire conditioned path, including all later observations.

Derivation. Given $N_T = 2k + 1$, the ordered switch times are uniform in $0 < s_1 < \cdots < s_{2k+1} < T$. Their $2k + 2$ spacings are uniform on the simplex of total length T. The total positive duration A has density

$$f_k(a) = \frac{(2k+1)!}{(k!)^2 T^{2k+1}} a^k (T - a)^k, \qquad 0 < a < T.$$

Since $X_T = u(2A - T)$, the position Jacobian is $da/dx = 1/(2u)$. Multiplication by the Poisson count weight gives the joint density

$$\left.\frac{\Pr(X_T \in dx, N_T = 2k+1)}{dx}\right|_{x=0} = \frac{e^{-\lambda T}\lambda^{2k+1}}{2u(k!)^2}(T/2)^{2k}.$$

Its sum is $e^{-\lambda T}\lambda I_0(z)/(2u) > 0$. Division gives w_k. Conditioning the uniform spacings on $A = T/2$ leaves the two independent simplexes stated above. The occupation-time backbone is the equal-rate specialization of Cinque's Theorem 2.1, (2.6), printed p. 4; the source also gives the alternating switch representation (2.4), p. 3.

3. Midpoint law, including the atom

The exact midpoint law is an explicit pushforward of the preceding mixture. Let $d = (P_0, M_0, \ldots, P_k, M_k)$, $s_0 = 0$ and $s_j = \sum_{i<j} d_i$. Then

$$Y = X_{T/2} = u \sum_{j=0}^{2k+1} (-1)^j \min\{d_j, \max(0, T/2 - s_j)\}.$$

For any bounded measurable g, its expectation is $\sum_k w_k \mathbb{E}_k[g(Y)]$, where $\mathbb{E}_k$ is integration over the two normalized simplexes. This formula sums over midpoint velocities. The joint law retains $V_{T/2} = u(-1)^j$ on $s_j \le T/2 < s_{j+1}$.

In particular,

$$\boxed{\Pr(Y = uT/2) = \frac{1}{I_0(\lambda T)}.}$$

For $K = 0$ the unique switch occurs at $T/2$, and $Y = uT/2$ with right-continuous velocity $-u$. For $k \ge 1$, all simplex coordinates are strictly positive almost surely. Attaining $Y = uT/2$ would use all positive duration before the midpoint, leaving extra positive intervals with zero length. Attaining $Y = -uT/2$ would require zero initial positive duration. Both are simplex boundary events. The midpoint lies in an interval with index $1 \le j \le 2k$. For odd j its position is $u(2\sum_{i<j,\ i \text{ even}} d_i - T/2)$; for even j it is $u(T/2 - 2\sum_{i<j,\ i \text{ odd}} d_i)$. Each sum is a nonempty proper prefix of its simplex vector, so on each such region Y is a nonconstant affine function of the free simplex coordinates. Thus the $k \ge 1$ contribution is absolutely continuous on $(-uT/2, uT/2)$.

The velocity convention matters at the atom. On the one-switch component, conditioning instead on $|X_T| < \epsilon$ makes the switch time uniform in a symmetric interval around $T/2$. Half of these paths still have velocity $+u$ at the midpoint and half have $-u$. As $\epsilon \downarrow 0$, their position paths converge uniformly to the one-switch return path, but their midpoint velocity laws retain that equal mixture. Evaluation of velocity at a jump is discontinuous. Our exact bridge uses the right-continuous path version above; an endpoint-window measurement is a separate specified protocol.

4. Cuts sample one preparation

Every finite set of cuts is evaluated on this same count/simplex probability space. If ρ refines π, deleting the extra coordinates of $(X_t, V_t)_{t\in\rho}$ returns $(X_t, V_t)_{t\in\pi}$ pointwise. Hence their joint laws satisfy exact restriction consistency. This argument handles atoms without multiplying singular transition densities.

An independent midpoint-reset rule fails a direct test. Conditional on $Y = uT/2$, speed at most u forces velocity $+u$ almost everywhere on the first half and $-u$ almost everywhere on the second half. In particular $X_{T/4} = X_{3T/4} = uT/4$ deterministically. Giving either quarter point fresh nonzero variance changes the old preparation or violates its speed support. Velocity and the endpoint conditioning carry the required memory.

5. Exact action limit under arbitrary cuts

For any partition π of $[0, T]$, let X^π be the sampled linear interpolant, $|\pi|$ its largest interval, and $S_\pi = S[X^\pi]$. Then

$$S[X] = \frac{mu^2 T}{2}, \qquad 0 \le S[X] - S_\pi \le \frac{mu^2}{2} N_T |\pi|.$$

Indeed the polygon velocity on interval I is the average $\bar{V}_I$, and

$$S[X]-S_\pi=\frac{m}{2}\sum_{I\in\pi}\int_I (V_t-\bar V_I)^2dt.$$

An interval without an interior switch contributes zero. Every other interval contributes at most $mu^2|I|/2$; their total length is at most $N_T|\pi|$. The count is finite almost surely. Differentiating the convergent positive series for I_0, with $I_1=I_0'$, also gives

$$\mathbb{E}[N_T\mid \text{bridge}]=1+z\frac{I_1(z)}{I_0(z)}.$$

Consequently $S_\pi\to mu^2T/2$ almost surely and in L^1 on any deterministic shrinking meshes, with the displayed expected error bound. Also $\|X^\pi-X\|_\infty\le 2u|\pi|$. Along nested partitions S_π increases, by square completion at each inserted node.

The surviving action relative to the zero-position chord is the kinetic cost of the fixed returning path. The polygon approximation error tends to zero. Its value $mu^2T/2$ tends to zero with duration or speed. These are separate limits from A01's long-observation stationary coefficient $H_*=mu^2/\lambda$. For the bridge midpoint observable, C031 gives $0\le\kappa_{\text{mid}}=4m\operatorname{Var}(Y)/T\le mu^2T$. Its mean is generally biased, so midpoint action uses $2m\mathbb{E}[Y^2]/T$, rather than variance alone.

6. Midpoint crossover: observables

Keep the bridge preparation, count weights and velocity convention of §§1–3. Put $Q=2Y/(uT)$ and define $\kappa_{\text{mid}}=4m\operatorname{Var}(Y)/T$, $A_2=2m\mathbb{E}[Y^2]/T$, and the stationary reference plateau $H_*=mu^2/\lambda$. Cut refinement fixes the preparation; the window limit below varies T at fixed m,u,λ.

The exact crossover is

$$\kappa_{\text{mid}}/H_*=g(z)=zR(z)[1-R(z)],\qquad R(z)=\frac{\int_0^z I_0(s)\,ds}{zI_0(z)},\quad z=\lambda T.$$

It has cubic onset $g(z)=z^3/6+O(z^5)$ and tends to one for large z. The next sections derive the count-conditioned law, moments and mass-window constraint, using the same path construction throughout.

7. A beta law at fixed switch count

For $k=0$, $Q=1$ deterministically. For every integer $k\ge1$,

$$\boxed{\frac{Q+1}{2}\,\Big|\,K=k\ \sim\text{Beta}(k+1,k).}$$

Equivalently its density on $-1<q<1$ is

$$f_k(q)=\frac{(2k-1)!}{2^{2k-1}[(k-1)!]^2}(1+q)(1-q^2)^{k-1}.$$

Proof. For a telegraph segment of duration t, displacement x and initial velocity $+u$, let $a=(t+x/u)/2$, $b=(t-x/u)/2$. Uniform Poisson spacings, grouped into positive and negative durations, give joint position/count densities in the interior $|x|<ut$:

$$d_{2j+1}(x,t)=\frac{e^{-\lambda t}\lambda^{2j+1}}{2u(j!)^2}a^jb^j\quad(j\ge0),$$

$$d_{2j}(x,t)=\frac{e^{-\lambda t}\lambda^{2j}}{2u\,j!(j-1)!}a^jb^{j-1}\quad(j\ge1).$$

The zero-count law is an atom at ut, handled separately. For initial velocity $-u$, reverse the displacement sign. These formulas follow from Dirichlet spacings with respectively $(j+1, j+1)$ and $(j+1, j)$ groups; the position Jacobian is $1/(2u)$.

Split the return path at $t = T/2$ and take an interior midpoint y. The two possible midpoint velocity sectors have even/odd and odd/even counts. With $a = T(1+q)/4$, $b = T(1-q)/4$, each sector contributes $a^k b^{k-1}$ times a factorial sum. Their combined joint density in midpoint y and terminal position zero is

$$\frac{e^{-\lambda T}\lambda^{2k+1}}{2u^2}a^k b^{k-1} S_k, \qquad S_k = \sum_{j=1}^{k} \frac{1}{j!(j-1)![(k-j)!]^2} = \frac{k(2k)!}{2(k!)^4}.$$

For the last equality write each term as $j\binom{k}{j}^2/(k!)^2$, use symmetry $j \leftrightarrow k-j$ and Vandermonde's identity $\sum_j \binom{k}{j}^2 = \binom{2k}{k}$. Divide by C033's terminal position/count density $e^{-\lambda T}\lambda^{2k+1}(T/2)^{2k}/[2u(k!)^2]$ and multiply by $dy/dq = uT/2$. This yields f_k. It integrates to one, so at $k \geq 1$ the interior formula exhausts the conditional mass. At $k = 0$ retain the deterministic midpoint atom; its mixture mass is $1/I_0(z)$.

8. Exact moments and limits

Beta integration gives, also including the $k = 0$ atom,

$$\mathbb{E}[Q \mid K = k] = \mathbb{E}[Q^2 \mid K = k] = \frac{1}{2k+1}.$$

Summing the absolutely convergent series and integrating I_0 term by term therefore gives $\mathbb{E}Q = \mathbb{E}Q^2 = R(z)$. Consequently

$$\mathbb{E}Y = \frac{uT}{2}R(z), \qquad \mathrm{Var}(Y) = \frac{u^2T^2}{4}R(z)[1-R(z)],$$
$$\kappa_{\mathrm{mid}} = H_* g(z), \qquad A_2 = \frac{H_*}{2} z R(z).$$

For $z > 0$, $0 < R(z) < 1$, hence $g(z) > 0$. The exact identity also improves the generic midpoint bound to $g(z) \leq z/4$. The Taylor series gives $R(z) = 1 - z^2/6 + O(z^4)$ and thus the cubic onset in the opening result.

For the large-window limit use the integral representation $I_0(z) = \pi^{-1}\int_0^\pi e^{z\cos\theta} d\theta$, DLMF 10.32.1. The normalized exponential weight concentrates at $\theta = 0$: for any $0 < \delta < \pi$, the weight of $[\delta, \pi]$ is at most $(2\pi/\delta)e^{-z(\cos(\delta/2)-\cos\delta)}$. Differentiation under the integral then gives $I_0'(z)/I_0(z) \to 1$. Both $I_0(z)$ and $\int_0^z I_0(s)ds$ diverge, so l'Hopital's rule yields

$$zR(z) = \frac{\int_0^z I_0(s)ds}{I_0(z)} \longrightarrow 1, \qquad R(z) \longrightarrow 0, \qquad g(z) \longrightarrow 1.$$

At fixed m, u, λ, this proves

$$\kappa_{\mathrm{mid}}(T) \to H_*, \qquad A_2(T) \to H_*/2, \qquad \mathbb{E}Y \to \frac{u}{2\lambda} \quad (T \to \infty).$$

The normalized mean $\mathbb{E}Q$ vanishes; the unscaled midpoint mean retains a boundary bias. P02's sampler anticipated the plateau and cubic onset. This proof supplies the full curve and corrects its provisional statement that the unscaled midpoint mean tends to zero. Monotonicity of g is a separate question; the endpoint limits do not assume it.

9. Crossover windows and mass universality

For either C018's increment coefficient or C031's return-midpoint coefficient, a prescribed value $K_* > 0$ under a speed ceiling u can be realized only if

$$T \geq T_* := \frac{K_*}{mu^2}.$$

This follows by dividing $K_* \leq mu^2 T$ by the positive factor mu^2. For the present bridge the sharper $g \leq z/4$ gives $\kappa_{\mathrm{mid}} \leq mu^2 T/4$. These are necessary bounds on the observable, not sufficient conditions for any prescribed distribution.

A consequence for C035 is immediate. Suppose masses extend arbitrarily close to zero, the speed ceiling u and finite window T are common, and the increment or midpoint coefficient is the same $K_* \geq 0$ for every mass. Then $K_* \leq mu^2 T$ for all those masses forces $K_* = 0$. A positive mass-universal plateau is compatible with finite speed through mass-dependent crossover windows, rather than a common finite window valid for arbitrarily small masses.

For a telegraph family with shared plateau H_* and common speed u, $\lambda_m = mu^2/H_*$ and $T_{*,m} = 1/\lambda_m$ when $K_* = H_*$. The exact midpoint curve is $H_* g(mu^2 T/H_*)$. Its dependence on mT exhibits the nonuniformity of the large-window limit across small masses. The stationary increment instead uses $H_*[1 - (1 - e^{-2\lambda_m T})/(2\lambda_m T)]$.

At the additional formal identifications $H_* = \hbar$ and $u = c$, the scales are $T_{*,m} = \hbar/(mc^2)$ and $uT_{*,m} = \hbar/(mc)$, the reduced Compton time and length. The bounded-speed inequalities extend to $u = c$; the identification with a quantum phase scale belongs to the next dynamics test. This calculation concerns classical probability and specified windows.

10. Consequence and evidence record

The exact path law, consistent cuts, vanishing polygon error and positive long-window midpoint plateau concern different observables and limits of one preparation. Their combination supplies a concrete classical family for testing a proposed action-selection principle. The plateau value remains set by speed, mass and the supplied reversal rate.

C033–C034 retain the B14 path-law/cut audit. C037–C038 retain B17's midpoint and window audit; all proofs are preserved here. Cinque supplies the occupation law backbone, and the midpoint/action consequences are derived with the stated conditioning convention. Bounded-acceleration turns are developed in the mechanical return note; coherent composition has the separate premises of the checkerboard note.

This consolidated manuscript joins A06 and A09a. Historical check outputs remain in the repository; current verification uses written proofs and source review under the hard no-Python-numerical-verification rule.